Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/29/2014
PI Name:	German, Randall M. Ph.D.		
Project Title:	Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments (NNX14AB31G)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Del Mar	**State:**	CA
Zip Code:	92014	**Congressional District:**	53
Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	12/01/2013	**End Date:**	06/30/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Bhat, Biliyar	**Contact Phone:**	
Contact Email:	biliyar.n.bhat@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 6/30/2016 per NSSC information (Ed., 1/15/16) NOTE: End date is now 3/31/2016, per NSSC information (Ed., 9/9/15)		
Key Personnel Changes/Previous PI:	October 2014 report: Mr. Timothy Young, previously MS student on the project, has now returned as a part-time engineering researcher to assist in preparation of flight samples while being full-time employed as a quality engineer in the regional medical device industry. We are fortunate to have his guidance to ensure quality sample fabrication efforts. The postdoctoral researcher Dr. Jose A. Alvarado-Contreras, involved in computer simulations, has had to return to his university position and he has not been replaced as yet.		
COI Name (Institution):	Olevsky, Eugene (San Diego State University)		
Grant/Contract No.:	NNX14AB31G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (Ed., January 2014): Continuation of "Multi-Scale Modeling and Experimentation on Liquid Phase Sintering in Gravity and Microgravity Environments", grant # NNX10AV38G with the same Principal Investigator. A three year research project is proposed to investigate the gravity-dependent densification, microstructure development, and the related macroscopic shape evolution of multiphase powder materials during liquid phase sintering (LPS). The research is aimed at achieving two critical goals: (i) in-depth analysis of the LPS induced pore-grain structure evolution by the deconvolution of the gravity impact and (ii) exploring sintering under microgravity conditions as a technique for in-space fabrication and repair. Indeed, future NASA efforts to extend human exploration back to the Moon and beyond will require development of processes that permit fabrication under reduced gravity conditions. This capability is needed to reduce resource requirements for extraterrestrial construction and in space repair. Many proposed in-space fabrication and repair techniques involve particulate materials processing, such as via sintering, but limited data are available to assess the viability of these proposed net-shape techniques under reduced gravity. The research team will develop a computer code for the multiple scale simulation of sintering, embracing both microstructure development and macroscopic densification and associated shape distortion. Liquid phase sintering experiments on Earth and in microgravity will be conducted for a variety of heavy alloys to validate the sintering model framework. The anticipated research outcomes will be relevant to current and future space exploration needs for habitat creation, extraterrestrial exploration, and vehicle repair activities during trips to Mars or the Moon.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research is focused on liquid phase sintering, one of the most widely used manufacturing routes for ceramics, cemented carbides, powder metals, and composites. In the USA this is probably evaluated at $20 billion in annual product for applications that include components used in metal cutting, oil well drilling, automotive engines, motor bearings, computer heat sinks, jet engine valves, and even wrist watches (especially the newest cell phone coupled devices). These components are designed for high performance and high precision, and are often fabricated in large production quantities. The net-shape fabrication, where final size and shape is achieved without machining, is a challenge since often dimensions change 15 to 20% due to shrinkage in going from the powder to the dense component. Thus, computer models of the sintering process are vital to proper tool and process design. Further, to attain specific mechanical, thermal, electrical, or other properties requires further attention to microstructure control. This research provides a computational model that predicts final size and shape and with multiple scale model predicts properties. Further, with robust inclusion of gravitational gradients, liquid-solid migration, and substrate friction, the computer simulation will be useful for predicting component attributes over a range of gravitational conditions as required for eventual fabrication on the moon, planets, or microgravity. For example, in building structure on the lunar surface or repairing space vehicles.
Task Progress:	The project is slightly behind schedule since the flight date to the International Space Station has been delayed by about six months. The San Diego State University team remains focused on meeting all qualification criteria, fabrication of test samples, supporting vapor tests, and other necessary trials to validate the safety of the experiments. Final fabrication of the 49 flight and 49 ground samples, plus spares, is nearing completion and details associated with packaging and containment will be demonstrated in a ground-based furnace located at Marshall Space Flight Center during the latter portion of 2014, probably extending into early 2015.
Bibliography Type:	Description: (Last Updated: 03/27/2025)
Papers from Meeting Proceedings	German RM. "A Review on Lower Sintering Temperature Tungsten Alloys." Ninth International Conference on Tungsten, Refractory, and Hard Materials, in conjunction with the 2014 World Congress on Powder Metallurgy & Particulate Materials – PM2014, Orlando, Florida, May 18-22, 2014. In: Proceedings of the Ninth International Conference on In: Tungsten, Refractory & Hardmaterials. Princeton, NJ : Metal Powder Industries Federation, 2014. p. 368-382. ISBN No. 978-0-9819496-7-7. , May-2014
Papers from Meeting Proceedings	Bollina R, German RM. "Microstructural Evolution of 93 Wt. % Tungsten Heavy Alloy: A Quenching Study." Ninth International Conference on Tungsten, Refractory, and Hard Materials, in conjunction with the 2014 World Congress on Powder Metallurgy & Particulate Materials – PM2014, Orlando, Florida, May 18-22, 2014. In: Proceedings of the Ninth International Conference on In: Tungsten, Refractory & Hardmaterials. Princeton, NJ : Metal Powder Industries Federation, 2014. p. 400-408. ISBN No. 978-0-9819496-7-7., May-2014 , May-2014
Papers from Meeting Proceedings	German RM. "Identification of the Common Densification Pathway for Metal Powder Compaction, Sintering, and Pressure-Assisted Densification." World Congress on Powder Metallurgy, Orlando, FL, May 2014 Proceedings of the 2014 World Congress on Powder Metallurgy & Particulate Materials, May 18–22, Orlando, FL In: Advances in Powder Metallurgy and Particulate Materials 2014. Proceedings of the 2014 World Congress on Powder Metallurgy & Particulate Materials, vol. 1. Princeton, NJ : Metal Powder Industries Federation, 2014. p. 132-149. ISBN-13: 978-0-9853397-6-0 , Sep-2014